

Database Project Requirements

Business based Justification

ISM 6212

Soyeong Cha

Executive Summary

Shelters across the nation have an increasing overcrowding crisis that results in the euthanizing of thousands of dogs annually simply because there is no room at home for them. In addition to adoption and fostering programs being available, various reasons keep the potential adopter from taking one of these lovely dogs into his or her life, including concerns over money, unknown behavior, lack of time, and being uninformed on the responsible dog ownership issue.

Simultaneously, shelters themselves lack effective process management for the adoption, available dog tracking, adoption status tracking, and maintaining placed dogs with rightful owners.

Foster Dog Tracking and Adoption System is an evidence-based system intended to decrease adoption effort intensity, encourage optimal shelter function, and support greater community involvement in animal care. This system provides a centralized structure to track shelter dogs, identify them with suitable adopters, handle volunteers and donors, and facilitate adoption-related communication. Through automation, systematic reporting, and real-time tracking, this system removes inefficiencies in the adoption process, prevents shelter overcrowding, and maximizes overall success rates of dogs being placed into their forever homes.

Problem Statement

In spite of increased pet adoption awareness, there are still more dogs remaining in shelters for extended periods of time before being euthanized when shelters become too crowded. This is a result of several of the major issues deterring fostering and adoption.

One of the biggest adoption barriers is the financial cost of dog ownership. Prospective adopters are concerned about the long-term costs of food, vet care, grooming, and emergency medical conditions. Shelter dogs can also require additional medical procedures or behavior therapy, so adoption appears to be an expensive undertaking. Without money or cheap veterinary care, many are hesitant to adopt.

One serious concern is a shelter dog's unknown behavior. Adopters are concerned that the dog will fail to fit in with their household, be unsuitable with existing pets, or have issues like separation anxiety and aggression. Given that so many shelter dogs were abused or neglected, they must be formally trained and socialized, and many adopters are not comfortable doing this.

Inefficient shelter management and adoption matching contribute to the process complexity. Shelters track available dogs and adopter information manually for most shelters, causing inefficiencies and lost adoption opportunities. There is a possibility of longer adoption request completions, and shelters will not be able to monitor pending adoptions, placements completed, and returned dogs without an automated system.

Shelter crowding and shortages of space contribute to the crisis as well. Shelters regularly operate at or above capacity but lack space-tracking capability to enable them to adjust intake levels meaningfully. The shelters are subsequently forced to exercise difficult decisions with respect to euthanasia despite the presence of adopters.

Besides, there is little community involved in animal welfare efforts. Shelters rely on foster homes, donors, and volunteers but lack an organized method to enable donations, track volunteer hours, and communicate with potential fosters. Without organized outreach, shelters have no chance of building support systems.

Foster Dog Tracking and Adoption System

To address these, the Foster Dog Tracking and Adoption System provides an end-to-end, database-driven solution improving shelter effectiveness, adoption rate, and involving community engagement.

One of the basic elements of this system is automated dog-adopter matching, and it checks adopter requirements (size of dog, breed, type of home, level of experience) against available dogs using a formatted score-matching algorithm. Adopters are helped to discover a dog appropriate to their lifestyle and reduces mismatched adoption and re-return.

The system also includes real-time monitoring of shelter capacity, which monitors space availability and alerts when shelters reach 70% or higher capacity. This allows shelters to forecast intake levels, adjust foster placements, and avoid overcrowding in advance.

A centralized adoption management system streamlines shelter operations by tracking the status of every dog, recording adopter information, and tracking the progress of adoption requests. Upon finalization of an adoption, the system automatically updates the status of the dog, preventing double postings and ensuring correct records.

The system also has volunteer and donation management capabilities so shelters can track foster applications, volunteer schedules, and financial and supply donations. The feature allows for more community engagement and provides shelters with the tools to efficiently operate the shelters.

Sophisticated reporting and analysis capabilities allow shelters to report on adoption patterns, shelter capacity, donation activity, and volunteer involvement. These statistics allow shelters to analyze operations, make data-driven decisions, and improve adoption success rates.

Business Impact

The Foster Dog Tracking and Adoption System will create long-term benefits for animal shelters, adopters, and the community as a whole by improving adoption rates, operational effectiveness, and information exchange between shelters and the public.

One of the most profound impacts of this system is the increase in adoption success rates. By using formalized adoption matching and computerized monitoring, shelters can quickly and accurately match adopters with suitable dogs, reducing mismatched adoptions and getting more dogs into homes where they stay forever. The speed with which the system handles applications, tracks adoption status, and facilitates communication ensures that shelters can better process adoptions at high volumes, increasing placement rates and decreasing shelter returns.

The system is also instrumental in preventing shelter overcrowding. With real-time monitoring of the capacity of the shelters, organizations can better regulate intake levels, allocate resources, and avoid filling up to maximum capacity. The preventive action reduces the number of emergency euthanasias caused by space limitations.

Also, automatic shelter records and adoption tracking significantly reduce the administrative workload from shelter staff. By eliminating manual data entry, improving record integrity, and automatic adoption status notifications, the system allows shelters to devote more time to front-line animal care than paperwork.

One of the most important advantages of the system is increased community engagement and support for animal welfare. Through the provision of a systematic environment for volunteers, fosters, and donors, shelters can expand their base of support and engage an active community. The system's donation and sponsorship tracking abilities help shelters to manage financial contributions, allocate resources efficiently, and increase long-term sustainability.

Long-Term Sustainability and Future Enhancements

To enhance the continuation of success and expansion of the system, the future upgrades shall include:

- Expansion of the database with additional data fields such as behavior testing, check-ins after adoption, and medical history tracking.

- Connection with internet adoption websites, where potential adopters can search for available dogs over the internet and apply for adoption from home.
- Adoption recommendation powered by AI, in which machine learning will analyze past patterns of adoption and suggest the most appropriate matches.
- Advanced resource and supply chain monitoring, enabling shelters to monitor food inventory, medical supplies, and vaccination schedules to streamline dog care and resource use.

With continuous system feature and data innovation, it will be an even more powerful tool for shelters, adopters, and community partners.

Conclusion

The Foster Dog Tracking and Adoption System is an innovative solution for shelter operations management, adoption process streamlining, and community engagement. By offering automated adoption matching, real-time tracking, systematic record-keeping, and report-level detail, the system addresses inefficiencies in pet adoption and allows more dogs to find their forever homes.

With AI-driven adoption matching, online integration, and enhanced shelter resource monitoring, the system will become even more efficient, sustainable, and effective in shelter adoption programs. By leveraging data-driven solutions, community outreach, and streamlined processes, this system is a giant leap towards ending overcrowding in shelters and increasing pet adoption success nationwide.